# Exit Ticket · reflect and retrieve

## Exit Ticket

Once a cached value is stored it is always correct.

## Use key vocabulary from memory

1. Explain input in your own words.

2. Explain output in your own words.

3. Explain precondition in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why omit cover colour from a simple book-loan model?

### Q2

What is a reusable subroutine?

### Q3

Calculate 7×7.

### Q4

Why is caching squares safer than caching changing attendance data?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

### Core idea 1

Define the contract — The input is an integer, and the output is its square; overflow is a possible precondition concern.

### Core idea 2

Trace repeated work — First call for 5 calculates and caches 25; second returns the stored result.

### Core idea 3

Evaluate caching — Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

**Key visual · complete concept**




### Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

## Question 2

Trace two calls of SquareCached(7) starting with an empty cache.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

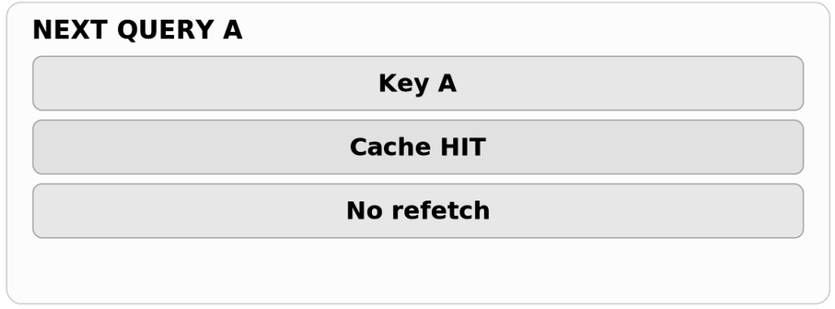
## Problem

A cached ticket price for zone A is £3.40. The same zone is requested twice.

## Follow the worked method

1. Predict first — The key A is already stored, so the lookup uses the cache.
2. Model and check — If pricing changes, the cached price must be updated or expired.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
Using the old cached 12 gives a stale answer instead of the new total 15.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.